Final Project
DS 4420 – Machine Learning 2

Northeastern University

Predicting Wildfire Severity and Occurrences using MLP and CNN models

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Purpose and Objectives

The purpose of this project is to utilize Multi-Layer Perceptron and Convolutional Neural Networks to predict wildfires and their severity.

MLP uses structured forest fire data containing FWI indices, temporal data, and location to be trained and tested on. It attempts to predict the area of forest fire damage in log-scaled hectares.

CNN uses RGB images of fire v. no-fire to be trained and tested on. It attempts to accurately predict whether an area will be impacted by fires or not.

Methodology & Model Architecture

Multilayer Perceptron (MLP):

- Two hidden layers with Leaky ReLU, L2 regularization, and batch normalization for stability and generalization.
- Trained using backpropagation and evaluated with MAE, RMSE, R^2, and MSE.

Convolutional Neural Network (CNN):

- Classifies 32x32 RGB images into fire vs. no fire using two convolutional layers (32 & 64 filters)
- Uses batch normalization, max pooling, and a dense layer for feature extraction and classification
- Final sigmoid-activated layer outputs binary probabilities; quantified with binary cross-entropy loss (for sigmoid)

Figure 1. A "fire image from "The Wildfire Dataset" [2].

Results and Discussion

MLP Model: Demonstrated strong learning on training data but failed to generalize (test $R^2 \approx 0$), indicating overfitting and architectural mismatch for the dataset size and quality. Additional regularization and improved feature engineering are needed for better test performance.

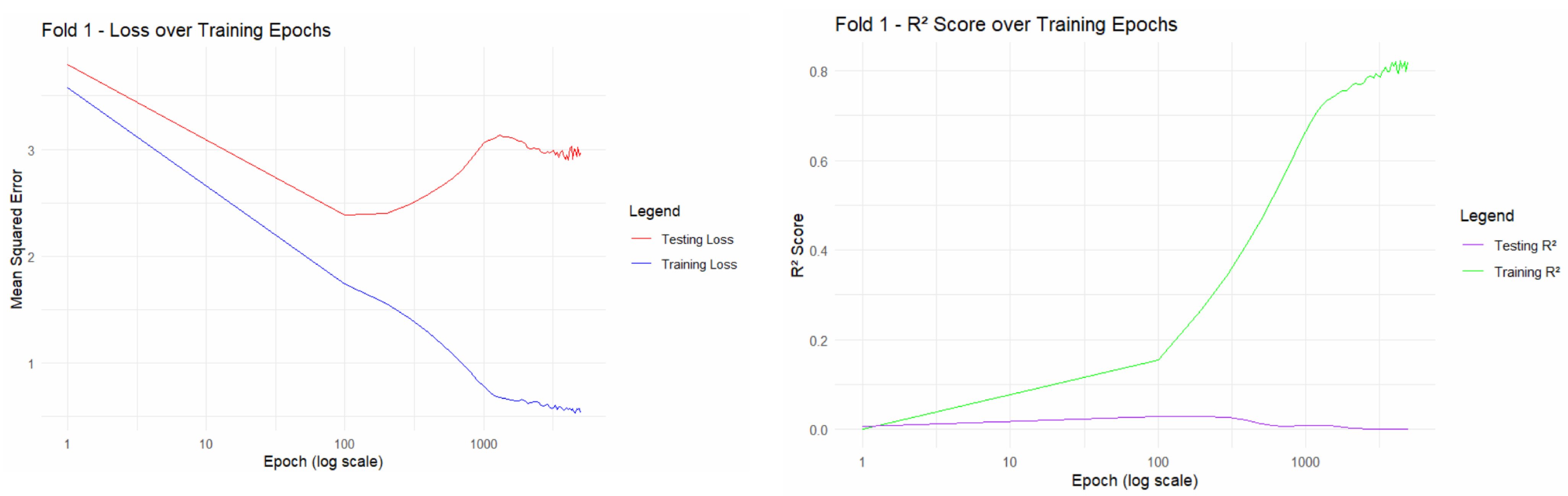

Figure 6: Loss and R^2 over epochs of training and testing data for MLP model.

Fold <int>	Train_Loss <dbl>	Test_Loss <dbl>	R2 <dbl>
1	0.5315677	2.963702	5.410092e-05
2	0.4839117	2.268476	2.966681e-02
3	0.5726035	3.919163	2.619177e-02
4	0.5532302	3.094592	1.899475e-04
5	0.5041858	2.419431	3.634765e-02

Figure 7: MLP metrics containing loss for training and testing sets and R^2 value.

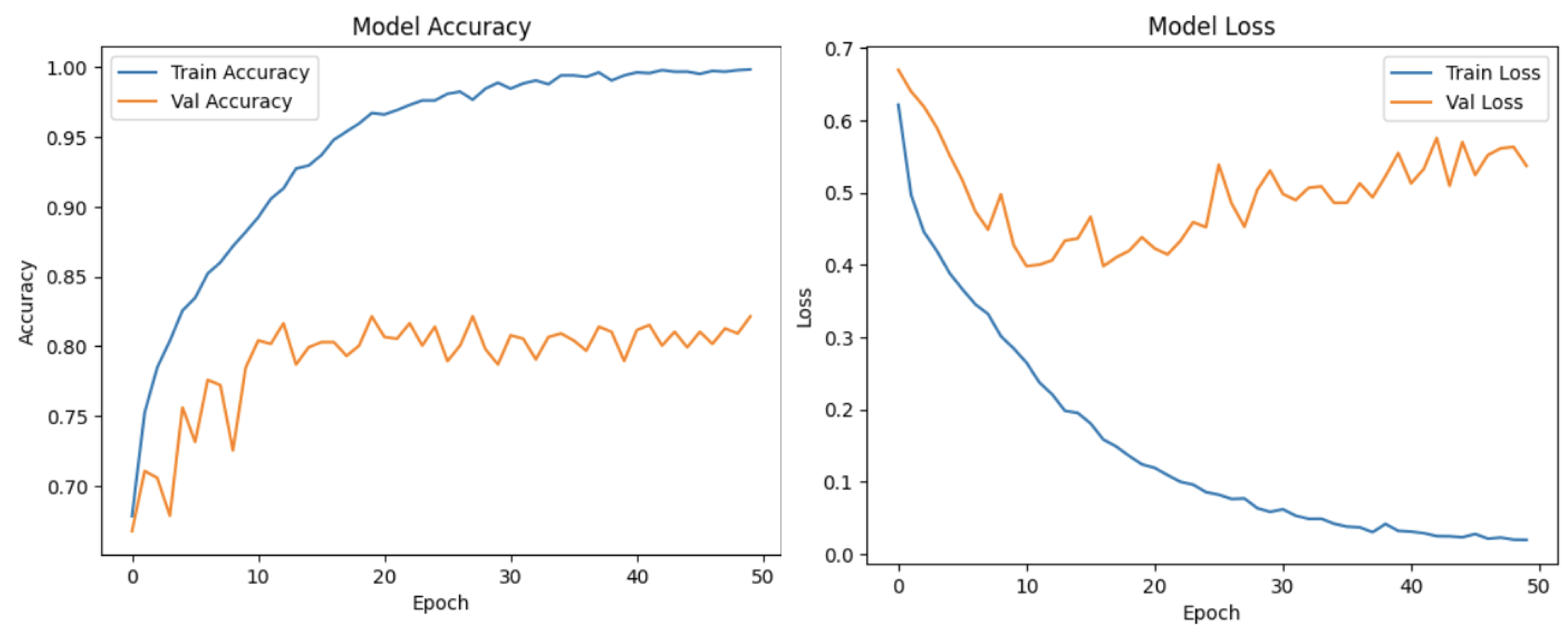

Figure 3 + 4: Accuracy and loss over epochs for training and validation sets for CNN.

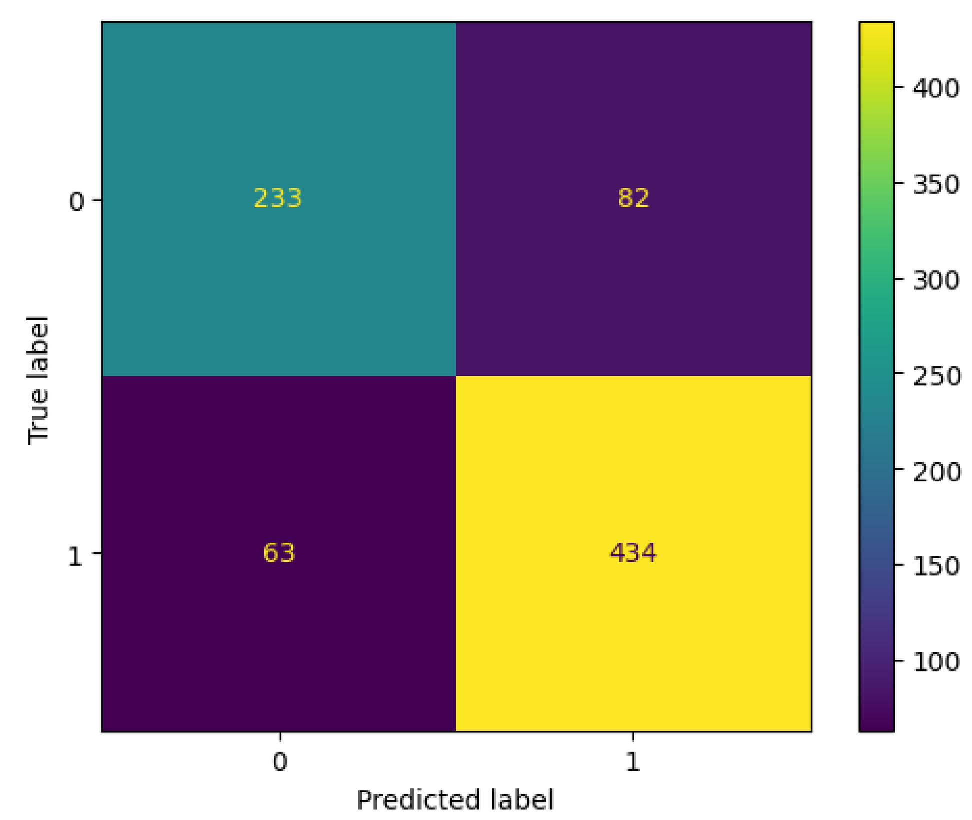

Figure 5: Confusion Matrix of the CNN model.

CNN Model: Achieved 82.14% validation accuracy, with moderate overfitting indicated by rising validation loss. Despite regularization the model misclassified some samples (82 false positives, 63 false negatives), suggesting potential gains from data augmentation or architecture tuning.

References

[1] Cortez, P, Morais, A. "A Data Mining Approach to Predict Forest Fires using Meteorological Data". In J. Neves, M. F. Santos and J. Machado Eds., New Trends in Artificial Intelligence, Proceedings of the 13th EPIA 2007 - Portuguese Conference on Artificial Intelligence, December, Guimaraes, Portugal, pp. 512-523, 2007. APPIA, ISBN-13 978-989-95618-0-9. Available at: http://www.dsi.uminho.pt/~pcortez/fires.pdf

[2] El-Madafri I, Peña M, Olmedo-Torre N. "The Wildfire Dataset: Enhancing Deep Learning-Based Forest Fire Detection with a Diverse Evolving Open-Source Dataset Focused on Data Representativeness and a Novel Multi-Task Learning Approach". Forests. 2023; 14(9):1697. https://doi.org/10.3390/f14091697